hwsynth: Natural Synthetic Handwriting in 153 Unicode Scripts from Writer, Movement, Tool and Scan Models

pavel9860
https://github.com/pavel9860/realistic_handwriting_153_scripts

October 2026

Abstract

Document understanding needs labelled handwriting in every script it meets. Learned handwriting generators cover a handful of scripts and need large per-script training sets; rendered handwriting fonts cover many scripts but look mechanical. We present hwsynth, a training-free generator that writes any text, including inline mathematics, chemistry and simple figures, in 153 of the 175 Unicode scripts. Letters are traced from handwriting fonts into pen paths. A writer model with 44 genes separates identity from situation and adds personal letter shapes and symbol habits. A movement model re-synthesises every stroke from lognormal sub-movements and trades shape fidelity against jerk energy by an effort parameter. A tool model with 17 instruments deposits ink by nib shape, ink class, speed and pressure, and a scan model ages and digitises the page. Twenty-four handwriting types can each be written in ten styles and with either hand; in addition, personas are fitted by analysis-by-synthesis to real writers of the IAM handwriting forms. A quality-diversity search fills a bank of realistic, stable and mutually distinguishable individuals per type. Evaluation uses real lines split by form so that no test form is used for fitting. On held-out IAM lines the searched bank reaches an embedding Fréchet distance of 0.139, closing 71 % of the gap between font rendering (0.415) and real handwriting of other forms (0.024); its 24 types are identified with 52 % top-1 accuracy (chance 4 %).

1 Introduction

The need. Handwriting is a large share of the documents that machines are asked to read: filled forms, letters, notes, prescriptions, exam sheets and archives. Text recognition, layout analysis and document dewarping need it in quantity, with exact labels (characters, words, boxes, pen paths), in every script the system will meet. Collecting and annotating real handwriting at that scale is slow and expensive, and for most of the world's scripts there is little or none.

The gap. Synthetic handwriting is the usual answer. The field has converged on learned generators: recurrent networks for pen trajectories [10], GANs [8, 15], transformers [2, 24] and diffusion models [22, 23] (survey: [5]). They are convincing *for the scripts and styles they were trained on*, mostly Latin, because they need thousands of labelled lines per script. For the other scripts practitioners render handwriting fonts with random affine jitter. The result is recognisably mechanical: every instance of a letter is the same glyph, strokes are uniform, baselines are straight, and the writing has no person behind it.

The question. Can a generator built from explicit models of *who* writes, *how the hand moves*, *what tool touches the paper* and *how the page is digitised* close that gap without a trained network, in every script that fonts cover? Two difficulties follow. Each model must remove a specific source of the mechanical look, and the dozens of interacting parameters must be set so that the output is realistic, stable and diverse, which hand-tuning does not achieve.

Contributions.

- A letter source for 153 scripts: centrelines traced from 59 handwriting fonts and 158 Noto fonts, drawn per grapheme cluster or per shaped word with correct bidirectional order (§4).
- A writer model that separates identity (the discriminating elements of forensic examination [14]) from situation, with 24 types, 10 styles, either hand, personal letter shapes and symbol habits (§5).
- A movement model joining lognormal sub-movements [25] with a minimum-jerk [7] effort trade-off that orders the styles from medical scrawl to calligraphy (§6).
- Tool, ink and scan models for 17 instruments and aged, digitised paper (§7), and inline mathematics, chemistry and figures (§8).
- Writer banks from a quality-diversity search and personas fitted to real writers (§10), evaluated against

real handwriting with form-disjoint splits (§9, §11).

Scope. hwsynth produces offline images of handwriting with labels; it does not recognise handwriting and does not imitate a specific person from a sample at inference time (personas are fitted offline). Realism is measured with proxies against real scans; a perception study is future work (§12). Code, fonts manifest, banks, personas and this paper are released under CC BY-NC 4.0.

2 Related Work

Learned handwriting generation. Graves [10] generated online trajectories with an LSTM and a mixture density output. Haines et al. [11] assembled a specific person's handwriting from their glyphs. ScrabbleGAN [8] and GANwriting [15] generate offline word images conditioned on text and style; Handwriting Transformers [2] and VATr [24] condition on a few style images; latent diffusion [22, 23] gives the most faithful styles today. All need labelled handwriting of the target script and style. Our generator needs none and covers 153 scripts, at the cost of the per-writer fidelity that learned models reach on their training scripts.

Fonts and perturbation. Before deep generators, synthetic training data came from geometric perturbation of real lines [32] and from fonts. Fonts scale to every script with a font but repeat glyphs exactly; we keep their coverage and redraw every glyph as a new movement by a person.

Models of handwriting movement. The Kinematic Theory [25, 26] represents a stroke as a sum of lognormal velocity impulses; its Sigma-Lognormal model reconstructs real trajectories [6] and has been used to design them [1]. Minimum-jerk planning [7] explains the smoothness of arm movements. We use the first for timing and corner cutting and the second for the trade-off between effort and shape.

Individuality and writer identification. Forensic examiners compare allograph design, connections, proportions, slant, spacing, baseline, commencements and terminations, line quality and natural variation [14]; Srihari et al. [30] quantified individuality. Automatic writer identification uses contour-direction and contour-hinge distributions [3, 27] and path signatures [4, 17, 19]. We use these features to measure and enforce that our types are distinguishable and to fit personas to real writers.

Ink, paper and scans. Franke and Rose [9] related pen force and ink deposition for solid, viscous and fluid inks; Sousa and Buchanan [29] modelled graphite on paper. Our tool and scan models follow these classes as renderers, not physical simulations.

text → scripts, bidi order, units, $math$ → **letters** (font centrelines) → **writer** (identity, style, personal letters, habits) → **hand** (lognormal sub-movements, minimum jerk, wrist pivots, tremor) → **tool** (width, density, texture) → **page** (paper, ruling, smudges) → **scan** (ageing, fading, bleed-through, scanner) + labels

Figure 1: The generation pipeline. No stage is learned; banks and personas (§10) only select genes.

Evaluation and search. We report Fréchet distances [13] and k-NN precision and recall [16] on ResNet18 [12] embeddings and stroke features. The writer banks are filled by MAP-Elites [21], a quality-diversity algorithm related to novelty search [18], with iso-line variation [33].

3 System Overview

Figure 1 shows the pipeline. Text is split into words, inline mathematics and drawing units. Each unit's glyph centreline is reshaped by the writer and placed on a line; the hand moves the pen over the strokes; the tool deposits ink; the page is composed on paper and optionally aged and scanned. Every stage is an explicit, inspectable model with its parameters in one configuration module. Writers are gene vectors; the banks of §10 only select genes, no stage is learned. Each sample comes with word boxes, the pen paths and the writer, tool and scan metadata.

4 Letters in Every Script

Scripts and units. Each character gets its Unicode script [31]. Scripts with independent letter forms (Latin, Cyrillic, Greek, Hebrew, Han, kana, Hangul, Ethiopic and 30 others) are drawn per grapheme cluster, so every letter receives its own variation; all other scripts are shaped by HarfBuzz/Raqm and drawn per word, so that Arabic joining and Indic conjuncts come from the font. Right-to-left runs are reversed while digits and Latin runs inside them keep their order. Scripts without spaces are split into short chunks (single characters for Han and kana), keeping closing punctuation with the preceding chunk.

Fonts. A manifest lists 59 handwriting fonts (SIL OFL or Apache) with class labels (cursive, print, brush), script restrictions and preferences (e.g. 80 % of Hebrew writers use a cursive Hebrew font), and 158 Noto Sans fonts as fallback. A font serves a script when it covers 90 % of the script's core letters (base letters without marks or compatibility forms; explicit alphabets for Latin, Cyrillic, Greek, Arabic, Han, kana, Hangul and Syriac). A writer uses its Latin font for every script that font serves and otherwise a handwriting font of the script; sixteen fonts

with a childish design are reserved for the child type and the uneducated style. Coverage: 153 scripts, 15 of them with handwriting fonts.

Centrelines. A glyph is rasterised at 160 px per em, skeletonised and traced into a graph: free spurs shorter than 5 % of the em are pruned, junction clusters merged, isolated dots kept, and edges continued through junctions along the smallest turn, which yields pen strokes in a natural order. Glyph size is normalised per script by the height of a reference letter in x-heights (e.g. Han twice the Latin x-height).

5 Writers

Genes. A writer is a vector of 44 genes in $[0, 1]$, decoded to natural units (Appendix, Table 4). Twenty-four are *identity* genes after the discriminating elements of [14]: size, width, ascender, descender and capital proportions, slant, letter and word spacing, line pitch, roundness and angularity (garland, arcade, angular and thread movement), joining, entry hooks and exit flicks, dot placement and form, pressure, the allograph set (font) and the share of letters in a second allograph, personal letter shapes, handedness, capitals-only writing and zone flattening. The others are *situational*: speed, effort, letter jitter, baseline wander, tremor, decay towards word ends, cross-outs and retracing.

Types and individuals. A type fixes a centre for every identity gene: listed values grounded in a description (Appendix, Table 5) and, for the rest, fixed values drawn from the type name, so that each type has one recognisable hand including its font. An individual draws each centred gene within ±12 % of its range. Genes the type leaves free are drawn from skewed priors: noise genes as u^k ($k = 2$–4), signed drifts near zero, rare events near zero. Uniform priors, used first, gave the median writer half of the maximum jitter on every channel at once, which looked unstable and childish; adults vary narrowly to moderately. The 24 types range from school cursive, copperplate and thread-like scripts through print, block capitals and micro writing to elderly, child and four rounded hands (rounded neat, bubbly print, looped cursive, soft upright).

Styles. A style is a situation applied to a person: situational genes are remapped into style ranges, keeping the person's relative position in each range, and a few identity genes are shifted (hurried writers join more and slant further). Ten styles are defined: casual, formal, accurate, calligraphic, professional, hurried, medical, educated, uneducated and unsupported (a sheet held in the hand). Each style also sets a tool prior and the paper.

Handedness. About 10 % of writers are left-handed (and all of the left-hooked type). They pull horizontal strokes from right to left, draw some loops clockwise, write more upright, hold broad nibs steeper and smear wet ink in the writing direction.

Natural variation. Repeated letters are never identical: neighbouring letters share correlated size, slant and baseline jitter (AR(1), $\rho = 0.6$); stroke ends overshoot or stop short; ink flow drifts from stroke to stroke; a fixed share of letters is written in a second allograph. Personal letter shapes give each writer its own version of every letter: a smooth deformation and a width, height and slant tweak drawn once per writer and letter from its identity, so that all instances of the writer's *a* share them while another writer's *a* differs. The identity is derived from genes that no style changes, so a person keeps its letters in every style.

Symbol habits. Writers differ in how they write symbols, which forensic examiners also compare. Each writer has habits drawn once from its identity, with probabilities set per type: crossed sevens and crossed *z*, slashed zeros, flagged ones, a quotation-mark style (English curly, straight, German low–high or French guillemets) and the plus sign for the ampersand. European cursive types cross sevens and *z* more often, drafting types slash zeros.

6 Hand Movement

Lognormal sub-movements. Each stroke is reduced to virtual targets (Douglas–Peucker turning points). Segment j becomes a sub-movement with velocity $\mathbf{D}_j \Lambda(t; t_{0j}, \mu_j, \sigma_j)$, where Λ is the lognormal impulse response of the Kinematic Theory [25]. Starts are spaced by $2\,o\,e^{\mu-\sigma^2}$, twice the overlap ratio o times the time of peak speed (isochrony). Fast, fluent writers overlap more and cut corners; slow, careful writers and discrete-stroke scripts (Han, kana, Hangul, Devanagari) reach every target. The speed of the synthesised movement drives the tool.

Effort and jerk. The integrated path is then the solution of

$$\min_{\mathbf{x}} \; \lambda(\epsilon)\,\|\mathbf{x}-\mathbf{p}\|^2+\|D_3\mathbf{x}\|^2, \qquad \log_{10}\lambda = -6+6.5\,\epsilon, \quad (1)$$

with D_3 the third difference (jerk) and end points held: a writer with low effort ϵ follows the minimum-jerk path and lets the shape go; with high effort the shape is kept at the cost of jerk energy. Effort is a situational gene whose style ranges rise from medical (0–0.12) through hurried, professional, casual, educated and formal to accurate and calligraphic (0.9–1).

Lines. Lines follow wrist-pivot arcs that reset at hand shifts every 25–60 mm, slow random baseline wander (correlation length 15–60 mm), an optional bow and slope, all damped on ruled paper; involuntary tremor is added last.

7 Tools, Ink, Paper and Scan

Tools. Seventeen tools combine a nib shape with an ink class [9] (Appendix, Table 6). Round nibs set width from the tip; broad nibs give the thick–thin contrast $w + e|\sin(\phi + \alpha)|$ for stroke direction ϕ and nib angle α; brushes widen with pressure$^{1.5}$ and taper. *Solid* ink (graphite) follows force and shows paper grain; *viscous* ink (ballpoint paste) thins with speed, skips on fast strokes and blobs where the pen lands; *fluid* inks (gel, rollerball, fineliner, fountain, markers) keep their density, pool at slow points, bleed and darken where strokes cross. Pressure combines the writer's push, its variation, extra force on downstrokes and lift-off ramps. Strokes are drawn as dense anti-aliased discs at 3× supersampling; the x-height is at least four nib widths.

Paper and scan. Paper has a tint, fibre noise and optional ruling that the writer follows. The scan model ages and digitises the page: yellowing and foxing with document age (Beta-distributed, mostly recent), ink fading, show-through of the reverse side, uneven light, greyscale or (rarely) bitonal conversion with an adaptive threshold, gamma, noise, blur, dust and JPEG compression. It is applied to half of all samples by default.

8 Mathematics, Chemistry and Figures

Inline `$...$` segments use a small TeX-like syntax: superscripts and subscripts, fractions, roots, Greek letters and symbols, drawn arrows, `\ce{...}` chemical equations (counts become subscripts, charges superscripts, reaction and equilibrium arrows) and `\benzene`. They are laid out as boxes of pen strokes: glyph runs come from the writer; rules, radicals, arrows and rings are drawn as strokes; the hand then moves over the whole expression like a word, so mathematics and chemistry match the surrounding writing. Figures (circle, ellipse, triangle, square, rectangle, arrow, star, tick, cross, axes with a curve, underline) are planned as freehand paths that do not quite close, overshoot at corners and bow slightly, and are moved and inked by the same models.

9 Real References and Leakage Control

IAM handwriting forms (primary). The handwritten part of 1 535 IAM forms [20] (one writer per form, clean scans, English) is cut into text lines (row-profile bands after removing rulings) at 150 dpi, giving 11397 lines from 1535 forms (7823 fit, 3574 test). The copy available to us has no writer identifiers, so the split is by form group: forms with the same text (e.g. a01-000, a01-000u, a01-000x) are always on the same side, and 30 % of the groups are held out as the test split. Writers who wrote several groups may appear on both sides; this is the one leakage we cannot exclude.

Handwritten web documents (secondary). Lines of 980 handwritten documents collected from the web, labelled by the language tag of their file names, give script targets beyond Latin (Cyrillic, Japanese, Hebrew, Greek, Arabic; a few documents each). They may contain printed lines and are used only for the per-script targets of Figure 2.

Stability from real handwriting. Baseline instability is the standard deviation of word ink centres around their fitted line, in x-heights (from the dense core band of the row profile). On clean real lines it runs from about 0.13 (steady) to 0.35 (wavy); higher values come from extraction errors. Real and generated lines are measured with the same function.

Leakage rules. Search and persona fitting use only the fit split; metrics use only the test split. Search, fitting and metrics use different texts. Bank admission is re-evaluated with fresh seeds. Realism statistics are normalised leave-one-document-out.

10 Writer Banks and Personas

Quality-diversity search. Each type has a MAP-Elites archive [21] over two measured behaviour descriptors: the slant of near-vertical pen-path segments (7 bins) and pen lifts per letter (5 bins). An individual is rendered on four search texts, in the casual style and a random other style, and scored by

$$Q = \mathit{realism} - 4\max(0,\ \mathit{instability} - 0.15) - [\mathit{separation} \le 0]. \tag{2}$$

Realism is the mean of two z-scores: the cosine similarity of the line embeddings to their five nearest fit lines, and minus the distance of the standardised stroke features to their five nearest fit lines. Each is standardised by the same statistic of real fit lines against other documents. *Separation* is the distance to the nearest other type's centroid minus the distance to the own centroid, in contour-hinge space [3] (χ^2) and path-signature space [17]. New individuals are iso-line variations [33] of two elites of the same type within the type's range. Elites are banked when a re-evaluation with two fresh seeds keeps realism at or above the median of all re-evaluated elites, separation > 0 and instability ≤ 0.35. The threshold is relative because realism is a z-score against the reference set: against the homogeneous IAM forms even good writers score below the real mean, and a fixed threshold calibrated on another reference rejects all.

Personas fitted to real writers. For each fit-split IAM form an evolution strategy (10 random starts of random types, 5 generations of 8 children, step size decaying from 0.12) searches 22 genes so that two generated lines match the writer: χ^2 distance of the contour-hinge distributions plus 0.02 times the mean squared difference of standardised stroke features, both averaged over the writer's lines. Each persona keeps its form, distance and nearest type, whose font class and symbol habits it uses. The fitting is resumable: every persona is saved when it is found. 203 personas were fitted (median distance 0.105, interquartile range 0.085–0.140); their nearest types are spread over 24 of the 24 types (most frequent: block capitals 19, soft upright 17, round print 16, left hooked 12, arcade 12).

11 Evaluation

Protocol. All metrics are computed on the IAM test split with texts disjoint from the search and fitting texts. Systems: rendering a handwriting font (*font*); uniform random genes (*random*); type priors (*prior*); the searched bank (*bank*); the fitted personas (*personas*); and the bank without the scan, pen, variation or joining models. Measures: realism (§10); Fréchet distance (FD) of ResNet18 embeddings and of stroke features [13]; k-NN precision and recall [16]; diversity (mean pairwise cosine distance of writer embeddings); writer identifiability (1-NN over lines); character error rate of Tesseract [28] on English lines; time per line. The reference row compares real test lines of one half of the forms with the other half.

Results. Table 1: the searched bank closes 71 % of the embedding-FD gap between font rendering and real handwriting and raises precision from 0.10 to 0.14. The fitted personas reach FD 0.129 and precision 0.12. Without the scan model FD rises to 0.242 and recall falls to 0.001, while realism, which treats scanner noise as atypical of the clean IAM scans, moves up. Removing variation, joins or the pen model lowers both realism and precision; without joins the embedding FD is slightly lower, so FD alone does not rank these ablations. Table 2: movement energy rises with effort from medical to calligraphic writing, entropy with it; accurate writing is the most legible for Tesseract and medical writing the least. Table 3: the search makes the types much easier to tell apart while individuals stay identifiable.

The path from mechanical to real. Figure 2 places real lines and generated lines in one plane: the first two principal components of real lines only (embeddings with standardised stroke features). Font rendering lies far from every real cloud. Each model moves the styles towards the real Latin cloud; the largest step is the writer model (redrawing glyphs as pen strokes); the later stages, most clearly the scan model, the searched bank and the fitted personas, move the styles further towards the centre of the IAM cloud. Calligraphy stays apart, as broad-nib contrast is rare in everyday handwriting. Text in other scripts leaves the font position and moves into the general handwriting region, but not reliably towards its own small real cloud (Arabic moves away from its fifteen real lines): the plane is dominated by Latin handwriting and the non-Latin targets are too small and too mixed to show script-specific convergence. Figure 3 shows the stages with the realism score: the writer stage is the large step, after which the score changes little.

Search dynamics. Figure 4: over 12 iterations the median elite quality rises from -2.61 to -1.81, median baseline instability stays at 0.15, within the range of clean real lines, and the archive fills 156 cells. The final re-evaluation (threshold: realism ≥ -2.18, the median) banks 54 individuals, 0–7 per type; types without banked individuals fall back to their priors.

12 Discussion and Limitations

What the measures can and cannot say. Realism is a proxy: ImageNet embeddings and 22 stroke statistics see texture, layout and stroke statistics, not whether letterforms are correct for a script, and the z-score treats scanner noise as atypical of the clean IAM scans, so the scan model improves FD and recall while lowering realism. FD and recall are the more informative measures for coverage of real handwriting. No measure replaces a perception study, where people try to tell real from generated lines.

References. The IAM copy has no writer identifiers, so test forms may share writers with fit forms. IAM is English and clean; the web documents add scripts but few documents each and possibly printed lines, so the non-Latin targets of Figure 2 are indicative.

Generator. Scripts without a handwriting font inherit Noto letterforms, which the writer and movement models make handwritten in execution but not in allograph design. Personas reproduce the statistics of a real writer's handwriting, not its exact letterforms. The banks are small after the strict re-evaluation. Generation takes 0.3–0.5 s per line on one CPU core.

Next steps. A perception study; handwriting fonts for more scripts; real references with writer identifiers and more scripts; page-level phenomena (margins, insertions, marginalia); learned realism judges; and using `hwsynth` as a teacher for learned generators in scripts without real data.

system	realism	FD emb.	FD feat.	precision	recall	diversity	writer id	CER	s/line
real (other forms)	–	0.024	2.93	0.717	0.634	–	–	–	–
font rendering	-5.247	0.415	228.90	0.100	0.001	0.0941	0.106	0.048	0.01
random writers	-1.984	0.152	21.89	0.081	0.350	0.1696	0.356	0.587	0.73
type priors	-2.159	0.146	22.05	0.069	0.303	0.1753	0.338	0.451	0.52
searched bank	-2.024	0.139	37.78	0.138	0.281	0.1722	0.231	0.417	0.37
fitted personas	-2.129	0.129	109.67	0.125	0.325	0.1705	0.263	0.390	0.30
bank, no scan model	-1.729	0.242	21.91	0.194	0.001	0.1215	0.356	0.363	0.35
bank, no pen model	-2.030	0.150	32.57	0.106	0.278	0.1739	0.294	0.393	0.29
bank, no variation	-2.041	0.150	32.46	0.087	0.401	0.1745	0.225	0.441	0.31
bank, no joins	-2.723	0.128	216.11	0.119	0.201	0.1728	0.331	0.374	0.30

Table 1: Systems against held-out IAM lines (casual style, texts disjoint from search and fitting). Lower FD and higher precision and recall are closer to real handwriting.

style	realism	FD emb.	energy	entropy	CER
medical	-2.355	0.189	1.6e-07	0.911	0.584
hurried	-2.462	0.165	3.2e-07	0.924	0.522
professional	-2.059	0.160	7.5e-07	0.961	0.370
casual	-2.024	0.139	1.6e-06	0.974	0.417
unsupported	-1.873	0.152	1.5e-06	0.970	0.441
educated	-1.889	0.165	3.7e-06	0.998	0.383
uneducated	-1.767	0.158	2.2e-06	0.978	0.234
formal	-2.239	0.164	6.8e-06	1.000	0.408
accurate	-1.776	0.177	9e-06	0.996	0.164
calligraphic	-2.578	0.158	3.4e-05	1.159	0.569

Table 2: Searched bank per style, ordered by effort: movement energy (jerk per unit length of the planned path), turning entropy (bits) and Tesseract CER.

writers	features	type	individual	style
prior	contour hinge	0.312	0.384	0.442
prior	path signature	0.292	0.542	0.484
bank	contour hinge	0.453	0.355	0.473
bank	path signature	0.522	0.447	0.541
chance		0.042	0.0104	0.200

Table 3: Top-1 identification on held-out texts (24 types, 5 styles): a type from other individuals, an individual from its other texts, a style from other individuals.

13 Conclusion

Explicit models of the writer, the hand, the tool and the scan, set by a leakage-controlled quality-diversity search and fitted to real writers, produce natural, stable and distinguishable handwriting in 153 scripts, with mathematics, chemistry and figures, without any trained generator. Every layer is inspectable and controllable: one can ask for a left-handed elderly writer in a hurry, with a pencil, on an old scanned page.

References

[1] Daniel Berio, Frederic Fol Leymarie, and Réjean Plamondon. Computer aided design of handwriting trajectories with the kinematic theory of rapid human movements. In *Conference of the International Graphonomics Society (IGS)*, 2017.

[2] Ankan Kumar Bhunia, Salman Khan, Hisham Cholakkal, Rao Muhammad Anwer, Fahad Shahbaz Khan, and Mubarak Shah. Handwriting transformers. In *ICCV*, 2021.

[3] Marius Bulacu and Lambert Schomaker. Text-independent writer identification and verification using textural and allographic features. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(4):701–717, 2007.

[4] Kuo-Tsai Chen. Integration of paths, geometric invariants and a generalized Baker-Hausdorff formula. *Annals of Mathematics*, 65(1):163–178, 1957.

[5] Moises Diaz, Andrea Mendoza-García, Miguel A. Ferrer, and Robert Sabourin. A survey of handwriting synthesis from 2019 to 2024: A comprehensive review. *Pattern Recognition*, 2025.

[6] Miguel A. Ferrer, Moises Diaz, Cristina Carmona-Duarte, and Réjean Plamondon. iDeLog: Iterative dual spatial and kinematic extraction of sigma-lognormal parameters. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(1):114–125, 2020.

[7] Tamar Flash and Neville Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7):1688–1703, 1985.

[8] Sharon Fogel, Hadar Averbuch-Elor, Sarel Cohen, Shai Mazor, and Roee Litman. ScrabbleGAN: Semi-supervised varying length handwritten text generation. In *CVPR*, 2020.

[9] Katrin Franke and Steffen Rose. Ink-deposition model: The relation of writing and ink deposition processes. In *International Workshop on Frontiers in Handwriting Recognition (IWFHR)*, 2004.

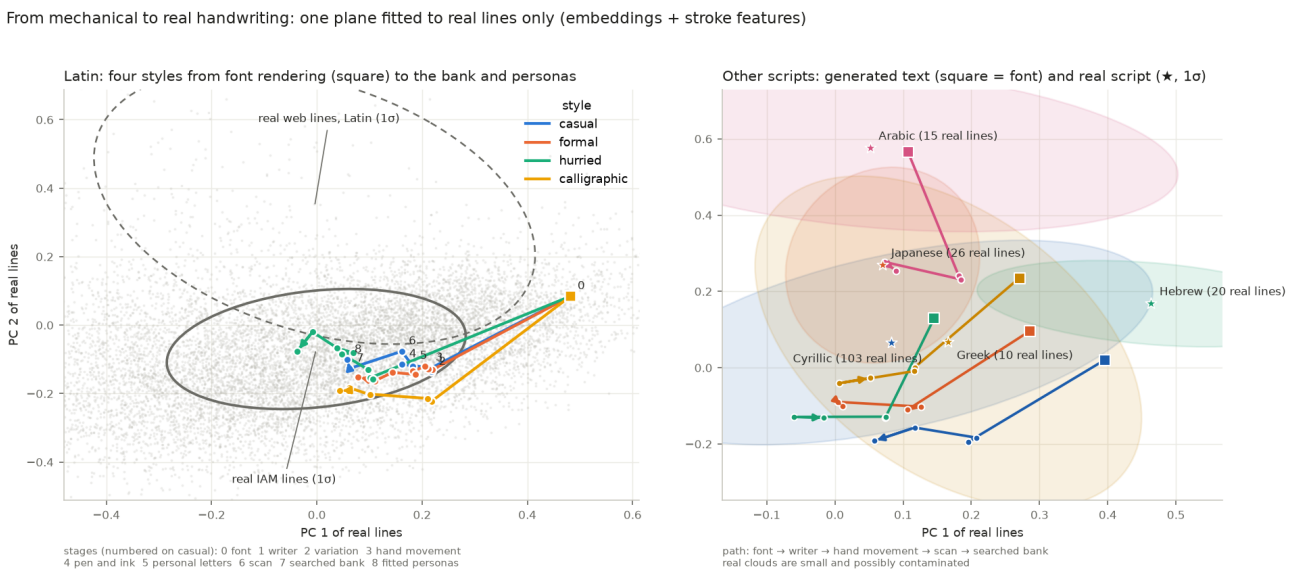


Figure 2: Handwriting in one plane fitted to real lines only. Clouds: real lines per script (IAM forms and web documents). Paths: generated Latin lines per style as the models are added, from font rendering to the searched bank and fitted personas (mean of 24 lines per point), and generated text of other scripts (casual style) from font rendering to the bank.

[10] Alex Graves. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.

[11] Tom S. F. Haines, Oisin Mac Aodha, and Gabriel J. Brostow. My text in your handwriting. *ACM Transactions on Graphics*, 35(3), 2016.

[12] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

[13] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *NeurIPS*, 2017.

[14] Roy A. Huber and A. M. Headrick. *Handwriting Identification: Facts and Fundamentals*. CRC Press, 1999.

[15] Lei Kang, Pau Riba, Yaxing Wang, Marçal Rusiñol, Alicia Fornés, and Mauricio Villegas. GANwriting: Content-conditioned generation of styled handwritten word images. In *ECCV*, 2020.

[16] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. In *NeurIPS*, 2019.

[17] Songxuan Lai and Lianwen Jin. Offline writer identification based on the path signature feature. In *ICDAR*, 2019.

[18] Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

[19] Terry J. Lyons. Differential equations driven by rough signals. *Revista Matemática Iberoamericana*, 14(2):215–310, 1998.

[20] U.-V. Marti and H. Bunke. The IAM-database: An English sentence database for offline handwriting recognition. *International Journal on Document Analysis and Recognition*, 5:39–46, 2002.

[21] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

[22] Konstantina Nikolaidou, George Retsinas, Vincent Christlein, Mathias Seuret, Giorgos Sfikas, Elisa Barney Smith, Hamam Mokayed, and Marcus Liwicki. WordStylist: Styled verbatim handwritten text generation with latent diffusion models. In *ICDAR*, 2023.

[23] Konstantina Nikolaidou, George Retsinas, Giorgos Sfikas, and Marcus Liwicki. DiffusionPen: Towards controlling the style of handwritten text generation. In *ECCV*, 2024.

[24] Vittorio Pippi, Silvia Cascianelli, and Rita Cucchiara. Handwritten text generation from visual archetypes. In *CVPR*, 2023.

[25] Réjean Plamondon. A kinematic theory of rapid human movements. Part I: Movement representation and generation. *Biological Cybernetics*, 72(4):295–307, 1995.

[26] Réjean Plamondon and Moussa Djioua. A multi-level representation paradigm for handwriting stroke generation. *Human Movement Science*, 25(4–5):586–607, 2006.

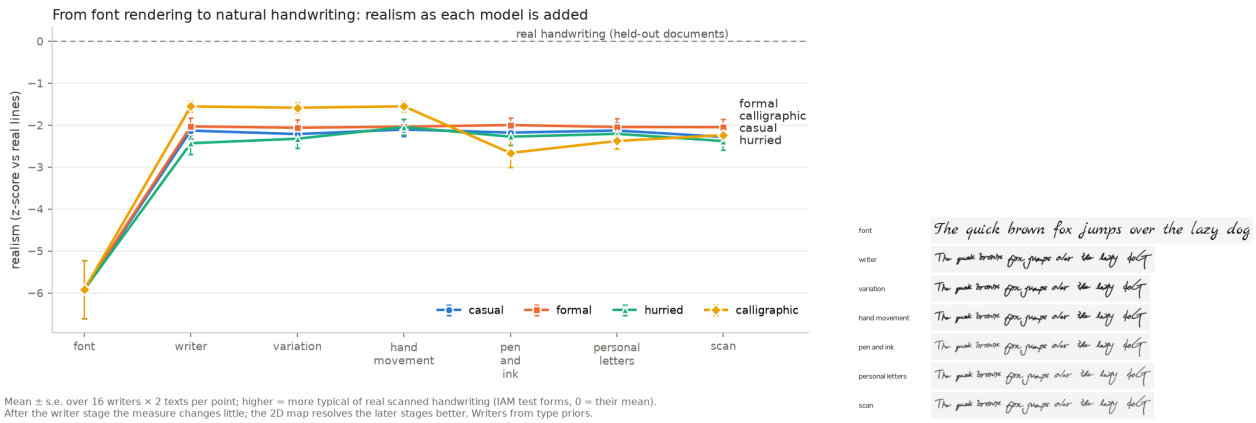


Figure 3: Realism on held-out IAM lines as each model is added (mean ± s.e., 16 writers × 2 texts per point; 0 is the mean of real lines), and one writer at each stage.

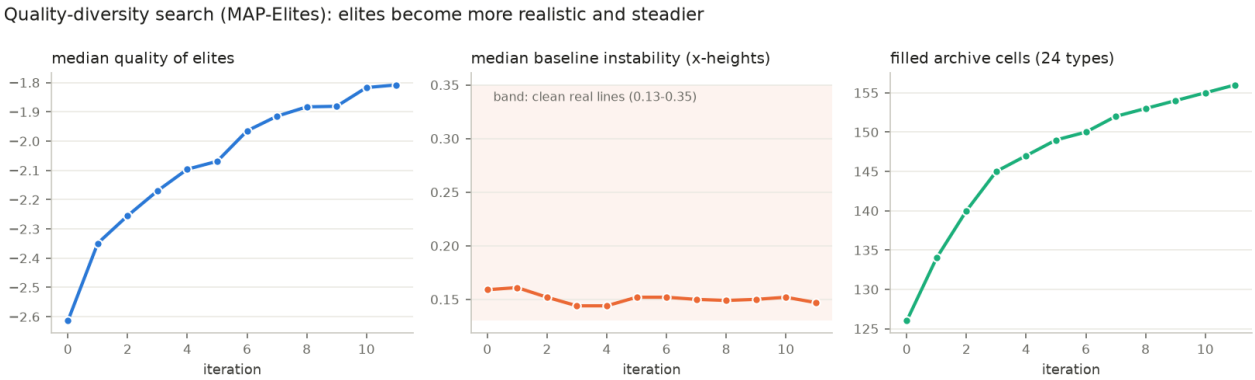


Figure 4: Quality-diversity search over 24 types: median elite quality, median baseline instability (band: clean real lines) and filled archive cells per iteration.

[27] Lambert Schomaker and Marius Bulacu. Automatic writer identification using connected-component contours and edge-based features of uppercase western script. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(6):787–798, 2004.

[28] Ray Smith. An overview of the Tesseract OCR engine. In *ICDAR*, 2007.

[29] Mario Costa Sousa and John W. Buchanan. Computer-generated graphite pencil rendering of 3D polygonal models. *Computer Graphics Forum*, 18(3):195–207, 1999.

[30] Sargur N. Srihari, Sung-Hyuk Cha, Hina Arora, and Sangjik Lee. Individuality of handwriting. *Journal of Forensic Sciences*, 47(4):856–872, 2002.

[31] The Unicode Consortium. The Unicode standard, annex #24: Unicode script property, 2025.

[32] Tamás Varga and Horst Bunke. Generation of synthetic training data for an HMM-based handwriting recognition system. In *ICDAR*, 2003.

[33] Vassilis Vassiliades and Jean-Baptiste Mouret. Discovering the elite hypervolume by leveraging interspecies correlation. In *Genetic and Evolutionary Computation Conference (GECCO)*, 2018.

A Genes, Types, Styles and Tools

gene	range	kind	meaning
size	1.8–3.4	identity	x-height (mm)
xs	0.75–1.3	identity	lateral expansion: horizontal scale of letters
asc	0.75–1.5	identity	ascender length (scale of the part above the x-height)
desc	0.75–1.5	identity	descender length (scale of the part below the baseline)
cap	0.9–1.5	identity	capital and first-letter size
slant	-15–32	identity	slant (deg, positive leans right)
slant_sd	0–4	situation	slant jitter per letter (deg)
size_sd	0–0.1	situation	size jitter per letter (share)
rot_sd	0–2.5	situation	rotation jitter per letter (deg)
base_sd	0–0.09	situation	baseline jitter per letter (x-heights)
ls	-0.1–0.3	identity	letter spacing (x-heights)
ls_sd	0–0.08	situation	letter spacing jitter (x-heights)
ws	0.7–2.2	identity	word spacing (x-heights)
pitch	2.5–4.2	identity	line pitch (x-heights) on blank paper
slope	-2.5–2.5	situation	line slope on blank paper (deg)
bow	-0.5–0.5	situation	line bow (x-heights)
wander	0–1	situation	slow baseline wander along the line
elastic	0–1	situation	smooth shape drift over the page
wobble	0–1	situation	low-frequency wobble along strokes
tremor	0–1	situation	high-frequency tremor
round	0–1	identity	corner rounding (garland, arcade movement)
angular	0–1	identity	corner sharpening (angular movement)
speed	0–1	situation	writing speed (rounding, ink skips, thin fast strokes)
effort	0–1	situation	movement effort: energy spent to keep the letter shapes (minimum jerk trade-off)
connect	0–1	identity	probability of joining letters
trail	0–1	situation	letters shrink and flatten towards word ends (thread movement)
flat	0–1	identity	vertical flattening of the x-height zone
hook	0–1	identity	probability of entry hooks (commencements)
flick	0–1	identity	probability of exit flicks (terminations)
dot_dx	-0.1–0.4	identity	dots and accents displaced in writing direction (x-heights)
dot_len	0–1	identity	dots drawn as dashes
ring	0–1	situation	dots drawn as small circles (above 0.5)
press	0–1	identity	mean pen pressure: light to strong push
press_sd	0–1	situation	pressure variation
down	0–1	situation	extra pressure on downstrokes
font	0–1	identity	allograph set: font choice
allo	0–0.4	identity	share of letters written in a second allograph
letters	0–1	situation	how far the writer's own letter shapes depart from the model letters
hand	0–1	identity	writing hand: left above LEFT_HANDED
cross	0–0.015	situation	probability of crossing out a misspelt word
retrace	0–0.03	situation	probability of retracing a stroke
caps	0–0.08	situation	probability of a capital letter inside a word
upper	0–1	identity	writes everything in capitals when above 0.5
drift	-0.25–0.25	situation	size drift over the text
margin_sd	0–1	situation	left margin wander (x-heights)

Table 4: The 45 writer genes.

type	fonts	habits	description
school_cursive	cursive	seven bar, z bar, one flag	taught joined script: loops, right slant, garlands
round_print	print	–	rounded upright print, open letters
compact_print	print	zero slash	small neat narrow print
block_capitals	print	zero slash	everything in capitals, drafting style
backhand	any	–	left-leaning script
angular_cursive	cursive	seven bar, z bar, one flag	sharp, narrow, strongly slanted joined script
thready	cursive	–	fast thread connections, letters dissolve
wide_garland	cursive	–	wide open letters, generous spacing
narrow_tall	any	–	narrow letters, long ascenders and descenders, tight spaci
micro	any	zero slash	very small writing
expansive	any	–	large writing, big capitals, wide spacing
arcade	cursive	–	rounded tops, slow upright joined script
print_script	any	–	print and script mixed
elderly	cursive	seven bar, z bar, one flag	slow tremulous joined script
child	print	–	large print, uneven baseline, heavy pressure
left_hooked	any	–	entry hooks, slight left slant, pressure on downstrokes
copperplate	cursive	seven bar, z bar, one flag	steep slant, long loops, ornate capitals
sloppy_fast	cursive	–	fluent but careless
ruled_neat	any	zero slash	very regular, even and aligned
rounded_neat	print	–	round, upright, regular and open: a rounded neat hand
bubbly_print	print	–	bubbly rounded print with circle dots
looped_cursive	cursive	seven bar, z bar, one flag	rounded joined script with full ascender and descender lo
soft_upright	any	–	soft upright hand, rounded joins, even size
artistic	cursive	–	embellished capitals, exaggerated loops and terminals

style	tools	paper	description
casual	all	any	everyday notes: as the person writes
formal	fountain fine, fountain medium, ballpoint	any	letters, applications: careful and even
accurate	ballpoint, gel, fineliner	any	forms and labels: separated careful letters
calligraphic	broad nib, italic nib, brush pen	any	broad-nib pen calligraphy: slow, contrasted
professional	ballpoint, gel, rollerball	any	fast, fluent, consistent office notes
hurried	ballpoint, ballpoint fine, gel	blank	written in a rush
medical	ballpoint, ballpoint fine, gel	blank	prescription scrawl
educated	ballpoint, fountain fine, fountain medium	any	fluent practised writing
uneducated	ballpoint, pencil 2b, pencil hb	any	slow unpractised print, mixed capitals
unsupported	ballpoint, gel, pencil hb	blank	on a sheet held in the hand, no support

Table 5: The 24 handwriting types and 10 styles.

B Examples

tool	nib	ink	width (mm)	edge (mm)	density
ballpoint_fine	round	viscous	0.22–0.3	–	0.75–0.9
ballpoint	round	viscous	0.28–0.4	–	0.78–0.92
ballpoint_bold	round	viscous	0.4–0.55	–	0.82–0.95
gel	round	fluid	0.3–0.55	–	0.9–1
rollerball	round	fluid	0.35–0.6	–	0.88–0.98
fineliner	round	fluid	0.18–0.35	–	0.9–1
felt_marker	round	fluid	0.6–1.1	–	0.8–0.95
chisel_marker	broad	fluid	0.25–0.4	1–2	0.75–0.9
fountain_fine	round	fluid	0.12–0.2	–	0.65–0.85
fountain_medium	broad	fluid	0.12–0.2	0.15–0.35	0.65–0.85
italic_nib	broad	fluid	0.06–0.1	0.5–0.9	0.75–0.9
broad_nib	broad	fluid	0.06–0.12	0.8–1.4	0.8–0.95
brush_pen	brush	fluid	0.15–0.3	–	0.85–0.98
pencil_hb	round	solid	0.3–0.5	–	0.45–0.6
pencil_2b	round	solid	0.35–0.6	–	0.6–0.78
pencil_2h	round	solid	0.25–0.4	–	0.4–0.55
mechanical_pencil	round	solid	0.2–0.3	–	0.45–0.62

Table 6: The 17 writing tools.

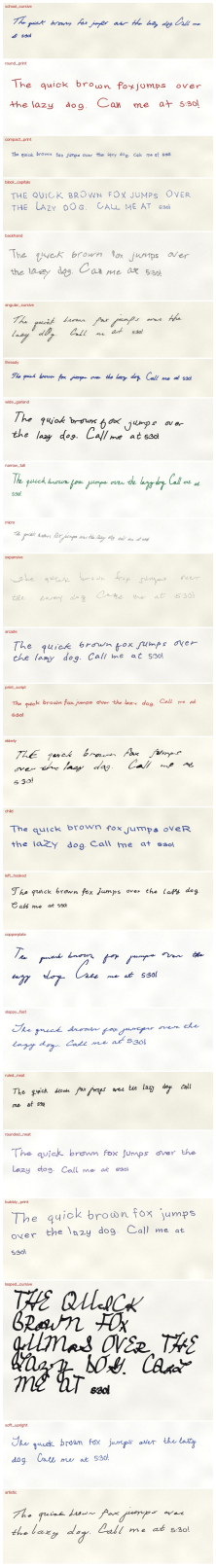


Figure 5: One individual of each of the 24 types (casual style).

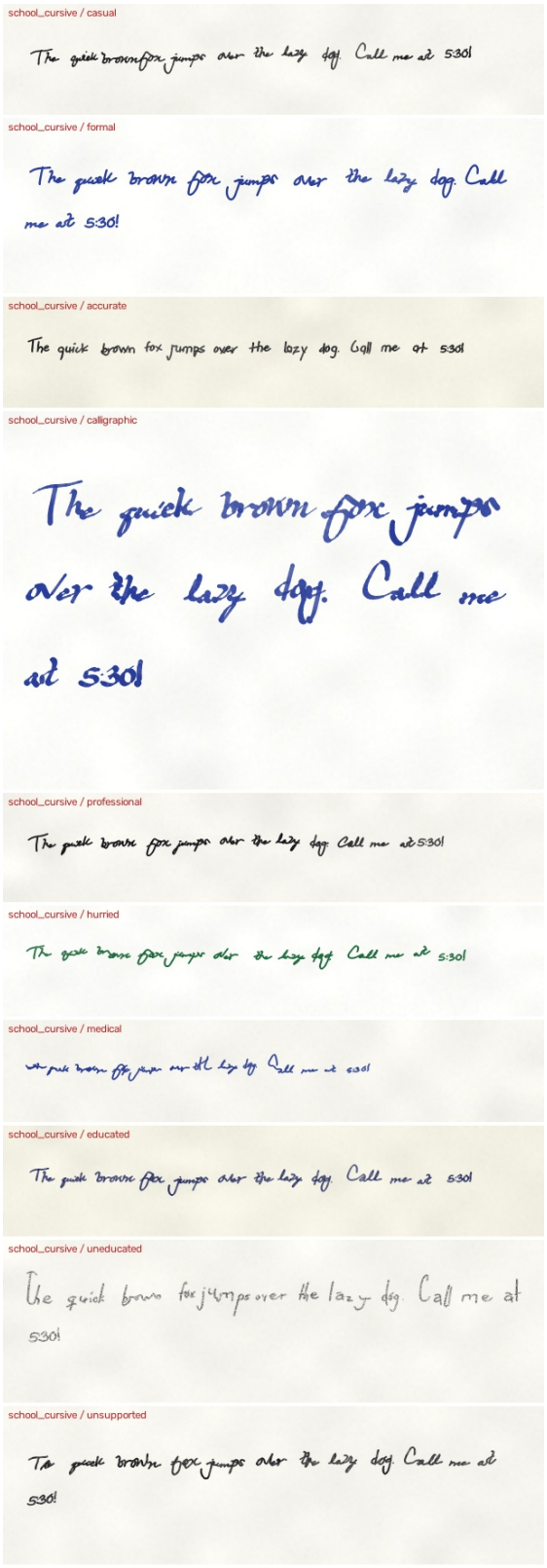


Figure 6: One school-cursive individual in the 10 styles.

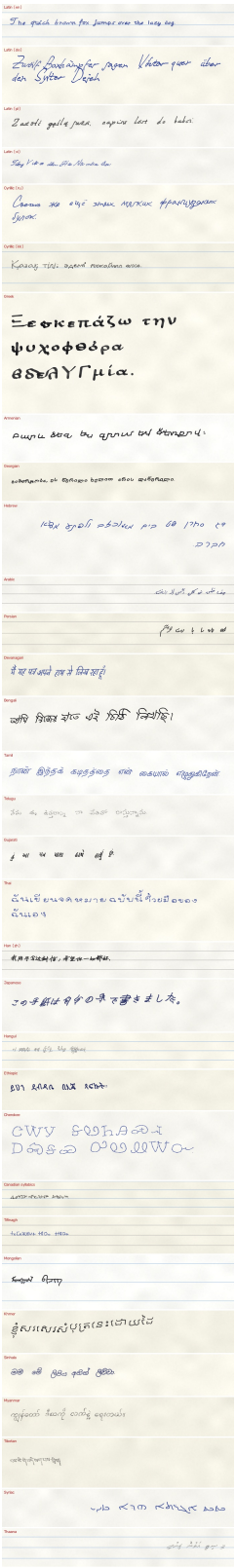


Figure 7: Random individuals writing 32 scripts.

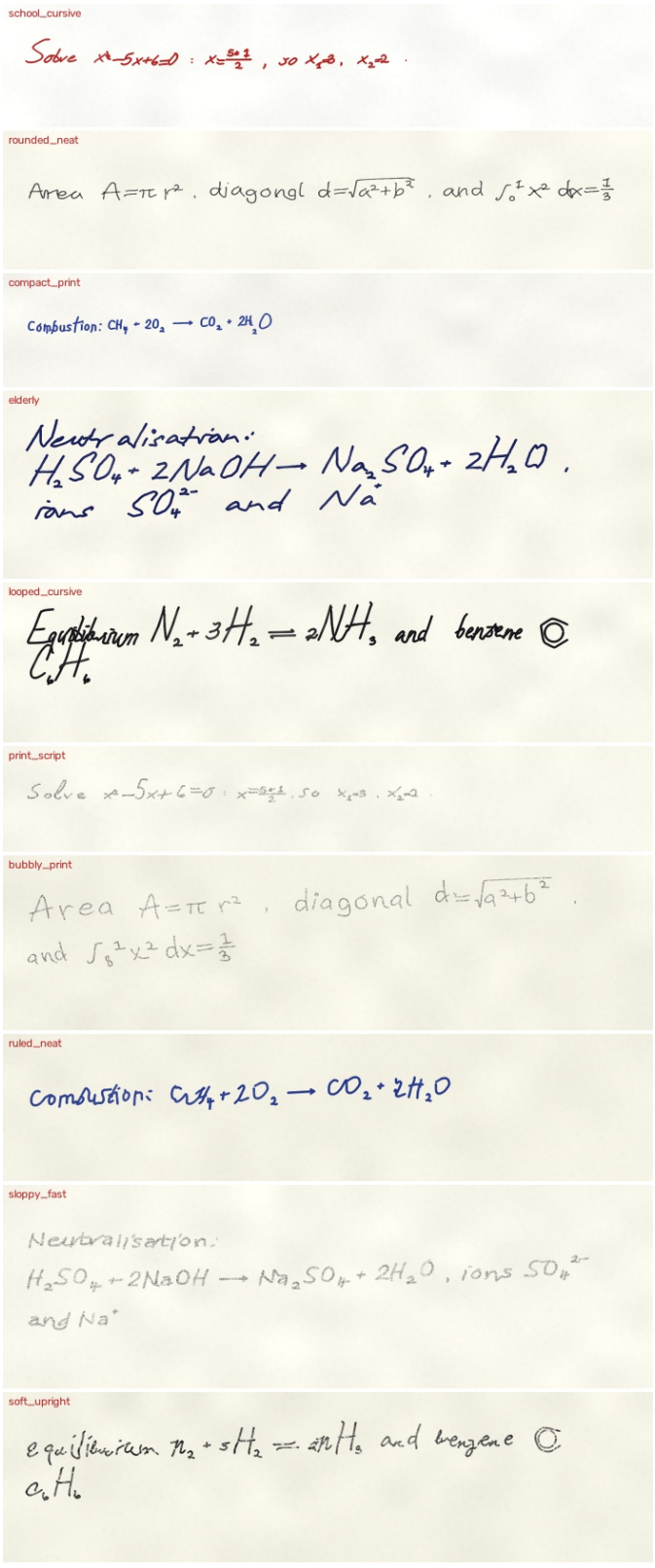


Figure 8: Inline mathematics and chemistry.

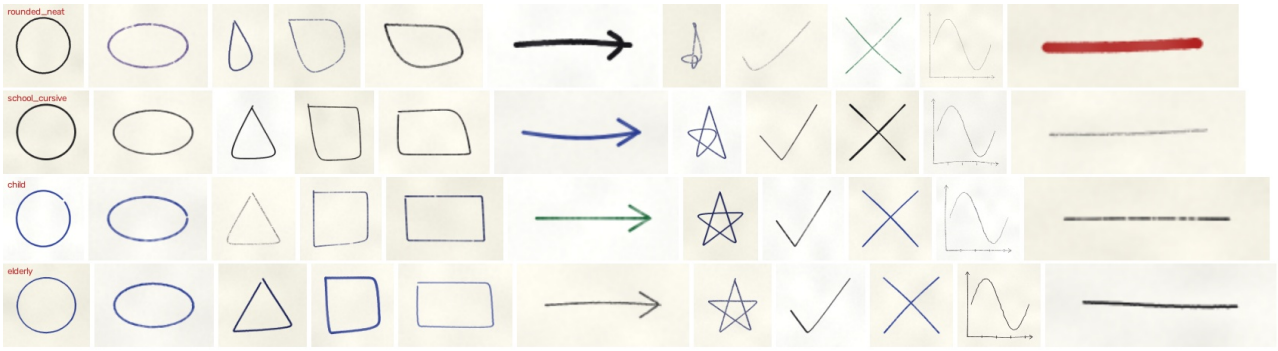


Figure 9: Hand-drawn figures by four types.



Figure 10: The 17 tools at light and strong push.



Figure 11: Right and left hand (gel, fountain pen, pencil): upright slant and smudges.